

Tuesday, 17/3/2026

Waves at Boundaries and Impedance Matching

1 *Characterize this Traveling Wave [Quick]

- (a) Given that the wavelength is $\lambda = 20\text{cm}$ for the wave which is moving in the positive x-direction. From the plot of displacement vs time, we can see that the amplitude of oscillations is 4cm. The plot also indicates that the time period is $\tau = 10\text{s}$. Then the frequency is $\nu = 0.1\text{Hz}$. Hence $\omega = 2\pi\nu = 0.2\pi/\text{s}$. The wavenumber $k = 2\pi/\lambda = 10\pi \text{ m}^{-1}$. The velocity of the wave is

$$v = \frac{\omega}{k} = \frac{0.2\pi/\text{s}}{10\pi \text{ m}^{-1}} = 0.02\text{m/s} \quad (1)$$

With all this information, we can say that

$$y(x, t) = 4\text{cm} \sin(10\pi x/\text{m} - 0.2\pi t/\text{s} + \phi_0) \quad (2)$$

Now from the plot, we see that at $t = 0, x = 0$, but then with increase in time t there is increase in the displacement. For the sine curve this can only happen if $\phi_0 = \pi$. So the displacement equation is now,

$$y(x, t) = 4\text{cm} \sin(10\pi x/\text{m} - 0.2\pi t/\text{s} + \pi) = 4\text{cm} \sin(0.2\pi t/\text{s} - 10\pi x/\text{m}) \quad (3)$$

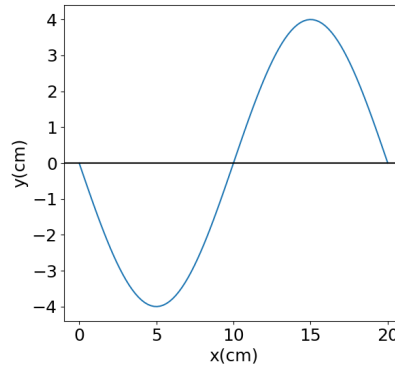


Figure 1: The plot showing dependence of y vs x at time $t = 0$.

- (b) The plot 1 shows the plot of y vs x at time $t = 0$.
(c) By differentiating the displacement equation, we find,

$$\dot{y}(x, t) = 4\text{cm} \times 0.2\pi \cos(0.2\pi t - 10\pi x) \quad (4)$$

$$\Rightarrow \dot{y}(0, 5\text{s}) = -4\text{cm} \times 0.2\pi = -0.8\pi\text{cm/s} = -2.513\text{cm/s}. \quad (5)$$

2 *Solving the Wave Boundary Conditions [Foundational]

- (a) Let the incident wave function be $y_i(x, t) = f_i(t - x/v_1)$, reflected be $f_r(t + x/v_1)$, transmitted be $f_t(t - x/v_2)$. Next, we shift the origin to be at the location of the massless knot. On each side of $x = 0$, we now have $y_L = y_i + y_r, y_R = y_t$.

Now, invoking continuity , we get,

$$f_i(0, t) + f_r(0, t) = f_t(0, t) \quad (6)$$

Next, invoking force balance so that knot doesn't accelerate to infinity,

$$T_1 \frac{\partial y_L}{\partial x} \Big|_{x=0} = T_2 \frac{\partial y_R}{\partial x} \Big|_{x=0} \quad (7)$$

This leads to

$$T_1(-f'_i(0, t)/v_1 + f'_r(0, t)/v_1) = T_2(-f'_t(0, t)/v_2) \implies Z_1(-f'_i(0, t) + f'_r(0, t)) = -Z_2 f'_t(0, t) \quad (8)$$

Now we can integrate out both sides w.r.t x to remove the derivatives (and the arguments are assumed to be $(x=0, t)$ for all functions) :

$$Z_1(-f_i + f_r) = -Z_2 f_t \quad (9)$$

Now, we make use of the Eq. 6 to get,

$$Z_1(f_r - f_i) = -Z_2(f_i + f_r) \implies f_r(Z_1 + Z_2) = f_i(Z_1 - Z_2) \quad (10)$$

Therefore, we get:

$$f_r(0, t)(Z_1 + Z_2) = f_i(0, t)(Z_1 - Z_2) \implies R \equiv \frac{f_r}{f_i} = \frac{Z_1 - Z_2}{Z_1 + Z_2} \quad (11)$$

Next, we also have $f_t = f_i + f_r$. So,

$$T = 1 + R = \frac{2Z_1}{Z_1 + Z_2} \quad (12)$$

Which is the standard result.

3 *Impedance Matching in Elastic Collisions [Substantive]

(a) Right away, we can invoke the conservation of energy, which in this case, is the kinetic energy:

$$\frac{1}{2}m_1v_i^2 = \frac{1}{2}m_1v_r^2 + \frac{1}{2}m_2v_t^2 \quad (13)$$

$$\implies m_1(v_i - v_r)(v_i + v_r) = m_2v_t^2 \quad (14)$$

Next, we use conservation of linear momentum to get:

$$m_1v_i = m_1v_r + m_2v_t \implies m_1(v_i - v_r) = m_2v_t \quad (15)$$

Dividing Eq. 14 by Eq. 15 we get:

$$v_t = v_i + v_r \quad (16)$$

Now, we can use this to eliminate v_t in the Eq. 15, to get:

$$m_1v_i = m_1v_r + m_2(v_i + v_r) \implies \boxed{\frac{v_r}{v_i} = \frac{m_1 - m_2}{m_1 + m_2}} \quad (17)$$

Next, we use Eq. 16 to get $v_t/v_i = 1 + v_r/v_i \implies \boxed{\frac{v_t}{v_i} = \frac{2m_1}{m_1 + m_2}}$.

(b) So, fraction of the transmission of kinetic energy is given by,

$$T \equiv \frac{KE_t}{KE_i} = \frac{\frac{1}{2}m_2v_t^2}{\frac{1}{2}m_1v_i^2} = \frac{m_2}{m_1} \frac{4m_1^2}{(m_1 + m_2)^2} = \frac{4m_1m_2}{(m_1 + m_2)^2} \quad (18)$$

If we define a number $\zeta \equiv m_1/m_2$, then $T = \frac{4\zeta}{(\zeta+1)^2}$

We can see that for both $\zeta \gg 1$ and $\zeta \ll 1$, this T reduces in value. Energy transfer is maximized if,

$$T' = \frac{4(1-\zeta)}{(1+\zeta)^3} \implies T(\zeta) = T_{max} \text{ if } \zeta = 1. \quad (19)$$

So if $m_1 = m_2$, T is maximized and the value is also 1.

(c) To compute the final fraction of KE transfer, we can simply multiply the Ts obtained:

$$T_{1,2} \times T_{2,3} \equiv \frac{E_2}{E_1} \times \frac{E_3}{E_2} = \frac{E_3}{E_1} = T_{1,3} \equiv T \quad (20)$$

So,

$$T = \frac{16m_1m_2^2m_3}{(m_1 + m_2)^2(m_2 + m_3)^2} \quad (21)$$

After taking log of T we get,

$$\ln(T) = \ln 16 + \ln m_1 + \ln m_3 + 2\ln m_2 - 2\ln(m_1 + m_2) - 2\ln(m_2 + m_3) \quad (22)$$

Taking the derivative and setting it to 0 we get,

$$\frac{1}{m_2} = \frac{1}{m_1 + m_2} + \frac{1}{m_2 + m_3} \implies m_1m_3 = m_2^2 \implies m_2 = \sqrt{m_1m_3} \quad (23)$$

- (d) All collisions happen pairwise, so the condition derived above is applicable here also. For $i \neq 1, N$

$$m_i = \sqrt{m_{i-1}m_{i+1}} \implies m_i^2 = m_{i-1}m_{i+1} \implies \frac{m_i}{m_{i-1}} = \frac{m_{i+1}}{m_i} \quad (24)$$

So, the ratios are constant. Let the ratio be r . Then,

$$m_2 = rm_1, m_3 = r^2m_1, \dots, m_N = r^{N-1}m_1. \implies r = \left(\frac{m_N}{m_1}\right)^{1/(N-1)} \quad (25)$$

$$m_n = m_1 r^{n-1} \implies \boxed{m_n = m_1 (m_N/m_1)^{\frac{n-1}{N-1}}}. \quad (26)$$

4 *Megaphones are Conical [Experiment]

Do the experiment.

5 Tsunami Shoaling: Waves in Shallow Water [Substantive]

- (a) So, from incompressibility of water, we get $\nabla \cdot \vec{v} = 0 \implies \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$.

Now, the velocity potential is given as,

$$\phi = A(z) \cos(kx - \omega t). \quad (27)$$

Now, $\frac{\partial^2 \phi}{\partial x^2} = -k^2 A(z) \cos(kx - \omega t)$ and $\frac{\partial^2 \phi}{\partial z^2} = A''(z) \cos(kx - \omega t)$.
Invoking incompressibility we get,

$$-k^2 A(z) \cos(kx - \omega t) = A''(z) \cos(kx - \omega t) \implies \boxed{A''(z) = k^2 A(z)} \quad (28)$$

- (b) The solution of the above equation for $A(z)$ is simply,

$$A(z) = C_1 e^{kz} + C_2 e^{-kz} \quad (29)$$

Now, as opposed to the last week's problem, this time, we find the z -velocity to be 0 at a finite depth.

So we invoke $v_z(z = -H) = 0 \implies \left. \frac{\partial \phi}{\partial z} \right|_{z=-H} = 0 \quad \forall x, t$

So now, we can write

$$C_1 k e^{kz} - C_2 k e^{-kz} \Big|_{-H} = 0 \implies C_1 e^{-kH} = C_2 e^{kH}. \quad (30)$$

Next, eliminate C_2 from the general solution,

$$A(z) = C_1 e^{kz} + C_1 e^{-2kH} e^{-kz} \quad (31)$$

$$= C_1 e^{-kH} e^{kH} e^{kz} + C_1 e^{-kH} e^{-kH} e^{-kz} \quad (32)$$

$$= C_1 e^{-kH} e^{k(z+H)} + C_1 e^{-kH} e^{-k(z+H)}. \quad (33)$$

$$\implies \boxed{A(z) = C \cosh(k(z+H))}, \text{ with } C = 2C_1 e^{-kH} \quad (34)$$

- (c) Vertical displacement η is given by $\eta(x, t) = B \sin(kx - \omega t)$.

Now, we have,

$$\frac{\partial \eta}{\partial t} = v_z(0) \implies -\omega B \cos(kx - \omega t) = A'(0) \cos(kx - \omega t) \quad (35)$$

$$= Ck \sinh(k(z+H)) \Big|_{z=0} \cos(kx - \omega t) \quad (36)$$

$$= Ck \sinh(kH) \cos(kx - \omega t). \quad (37)$$

But $A(0) = A_0 = C \cosh(kH)$. So we get,

$$-B\omega = Ck \sinh(kH) = Ck \tanh(kH) \times \cosh(kH) = kA_0 \tanh(kH) \implies \boxed{-B\omega = kA_0 \tanh(kH)} \quad (38)$$

(d) Next we use the Bernoulli equation $\frac{\partial \phi}{\partial t} + \eta g = 0$ at $z=0$. So, starting with LHS, we have ,

$$\left. \frac{\partial \phi}{\partial t} \right|_{z=0} = A(0) \times -\omega \sin(kx - \omega t) = -A_0 \omega \sin(kx - \omega t). \quad (39)$$

$$\implies Bg \sin(kx - \omega t) = -A_0 \omega \sin(kx - \omega t) \quad (40)$$

$$\implies -\frac{k}{\omega} A_0 \tanh(kH) g = -\omega A_0 \implies \boxed{\omega^2 = kg \tanh(kH)} \quad (41)$$

(e) For waves with extremely long wavelength (order of 100km), $\lambda \gg H \implies kH \ll 1$. So we can simply write $A(z) = C \cosh(kz + kH) \approx \cosh(kz)$.

So now,

$$\phi = C \cosh(kz) \cos(kx - \omega t) \quad (42)$$

Now, since k itself is very small, and $H \geq |z|$, so we can Taylor-expand it into,

$$\phi \approx C(1 + k^2 z^2 / (2!) + \dots) \cos(kx - \omega t) \quad (43)$$

$$\implies v_z \approx Ck^2 z \cos(kx - \omega t) \text{ to the leading term} \quad (44)$$

Differentiating it again, we get,

$$\frac{\partial^2 \phi}{\partial z^2} = \frac{\partial v_z}{\partial z} \approx Ck^2 \sim \frac{1}{\lambda^2} \approx 0 \quad (45)$$

So this means we can make use of the incompressibility equation,

$$\frac{\partial v_x}{\partial x} = \frac{\partial^2 \phi}{\partial x^2} = -\frac{\partial^2 \phi}{\partial z^2} \approx 0 \implies v_x = \text{const.} \quad (46)$$

This implies that the longer wavelength waves (can approach order of 100km) are colossal slabs of water slowly moving together. This unimaginable mass of water is one frightening aspect of tsunami.

(f) Again, in the shallow water limit, $\lambda \gg H$, so $kH \ll 1$, hence, we can write,

$$\tanh(kH) = \frac{e^{kH} - e^{-kH}}{e^{kH} + e^{-kH}} = \frac{(1 + kH + k^2 H^2 / 2 + \dots) - (1 - kH + k^2 H^2 / 2)}{(1 + kH + k^2 H^2 / 2 + \dots) + (1 - kH + k^2 H^2 / 2)} \approx \frac{2kH}{2} = kH \quad (47)$$

So, we can reduce the dispersion relation to,

$$\omega^2 \approx kg \times kH = k^2 gH. \quad (48)$$

(g) Phase velocity $v = \omega/k$. In the shallow water limit we have, $\omega \approx k\sqrt{gH}$, so, $v \approx \sqrt{gH}$. In the last week's tutorial, we had the deep water dispersion relation of $\omega = \sqrt{gk}$, there velocity was $v = \omega/k = \sqrt{g/k}$. The shallow water waves have seabed depth H dependent velocity, and so when the waves get near the shore, H reduces, and so does v . This is why tsunamis slow down near the coast.

(h) We can imagine the ocean as a series of thin slices perpendicular to the wave motion. Then each of the slices will have have a seabed depth $H(x)$ very slightly different from the previous one. So we can imagine a reflection and transmission at the interface between each slice. Next, the impedance is something like driving force/velocity. Here, the velocity $v \sim H^{1/2}$ reduces every slice very minutely. As we recall, the reflection coefficient,

$$R = \frac{Z_i - Z_{i+1}}{Z_i + Z_{i+1}} \quad (49)$$

Since the impedances offered by the slices are very nearly equal to each other in case of slowly varying $H(x)$, we can say $R \approx 0, T \approx 1$ and most of the energy propagates through.

- (i) The energy flux is given by $E \times v$ which is approximately constant as the waves progress towards the shore. Now, it is given that $E \approx B^2 \rho g$, which can be seen from the cyclic transfer of energy from kinetic to potential, and the potential maximum is achieved at amplitude B , and so $E \sim B^2$ as in Harmonic oscillator. This combined with $Ev = \text{const.}$ we get, $B^2 v \approx \text{const.}$
- (j) We already have the result $v \approx \sqrt{gH}$. So,

$$B^2 \sqrt{gH} = \text{const.} \implies B^2 \propto \frac{1}{\sqrt{H}} \implies B \propto H^{-1/4} \quad (50)$$

This means that as the waves approach the coast, the depth gives up, velocity slows down, and to compensate for the enormous energy, the height of the slab of water rises very high, bringing in death and destruction.
